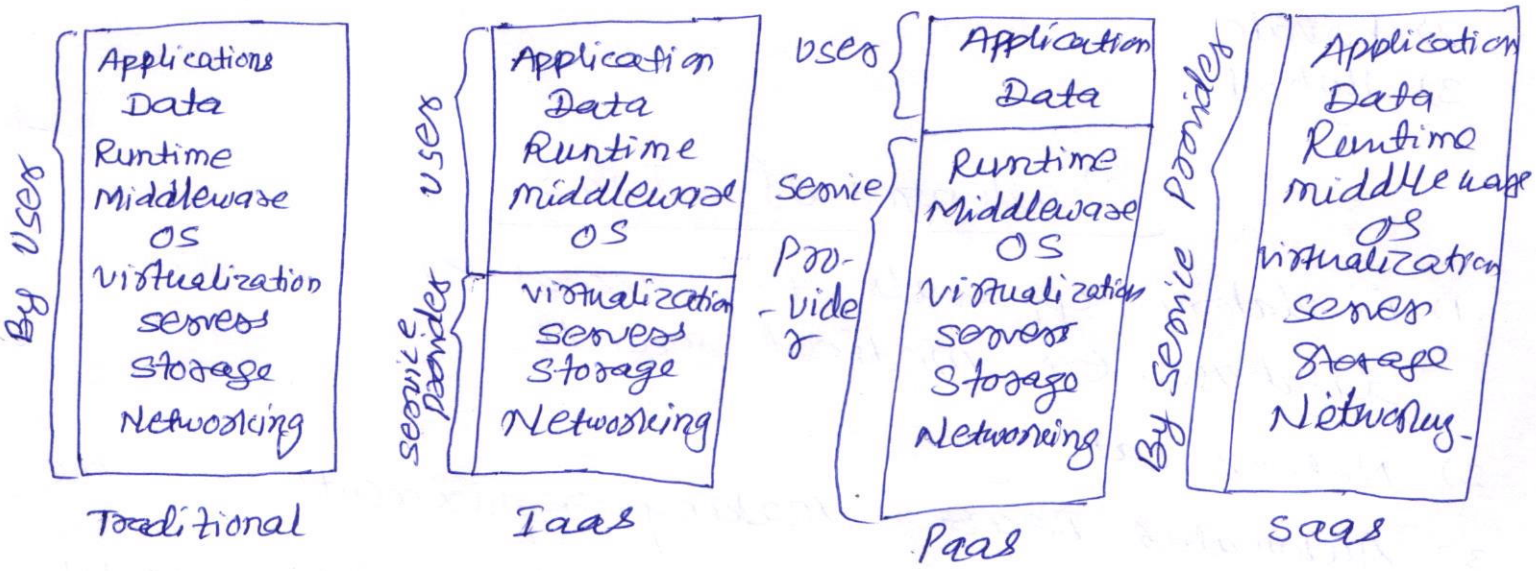
Service Models

Unit - IV



working Methodology of Iaas

- On-demand delivery of computing infrastructure
- Service provider provides virtualization of resources, servers, storage and networking equipments to users.
- Users rent servers, software, n/w equipment
- Service provider offers resource management

~~why Iaas~~ who opts Iaas Model

who opts

1) New businesses to start business without investment in hardware components.
2) Rapidly growing business
3) Business requiring fluctuating computing (Amazon, flipkart) during festival seasons.
4) Trials for various business models.

Essential Characteristics

1) Availability
2) Accessibility
3) manageability
4) performance
5) optimization
6) Reliability
7) portability

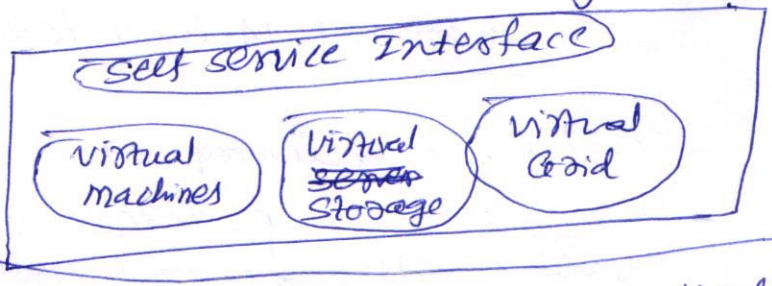
Deployment models preferred by Iaas

1) Public
2) Private
3) Hybrid

Challenges of Iaas

1) Regulatory approvals for access of sensitive database eg: medical records.
2) Network Lateney
3) Automated Decision making requirements by some users.
4) Seamless Scaling.

Iaas working Model



Paas

- Allows developers to use platform to develop applications.
- Allows deployment of created applications.
- Provides security to developers during development phase.
- Basically it is a middleware both. Cloud & developers.
- management of developed applications.
- Rent virtualized servers & services.
- Provides scaling.

working Model



Paas

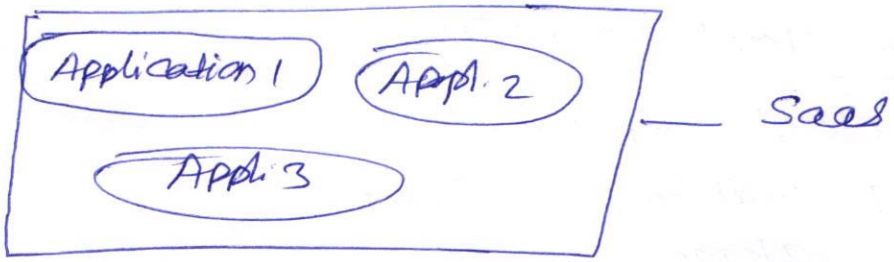
Saas

- Simplified software delivery model
- offered through thin clients
- fully pay-per-use model
- Remote access of software
- Access & control software via internet
- one to many model.

Essential characteristics of Saas

- Scalable for max. concurrency
- one to many model
- Transparency
- Customization of software
- seperate data space for different users.

working model



Challenges

- limited flexibility
- Users data security
- Centralized Control

Cloud service Management & Security

- Proper service management ensures optimal performance
- Ensure desired outcomes of customess
- Risk mitigation tech usage for seamless feeling to customes

Objectives

1) to provide standard services
2) to monitor usage & billing
3) to ensure high availability of n/w & connectivity
4) to fast allocation of requested resources
5) to provide secure computing & storage

Service Agreement

- Non-functional requirements expected from service provider.
- Clearly defined deliverables
- Quality & warranty of services
- Accounting & Billing
  - resource usage record as per metric
  - billing rules

security

Threats to Cloud computing

- Misuse of cloud computing
- Insecure application interfaces
- Data loss/leakage
- vulnerabilities due to sharing

Best Practices for Cloud Security

- Data Encryption
- Access control
- Multi-factor Authentication (MFA)